

C Programming Notes

Beginner-friendly study notes

1. What is C?

C is a general-purpose, procedural programming language. It is widely used for system programming, embedded systems, operating systems, and learning programming fundamentals.

2. Basic Structure of a C Program

```
#include <stdio.h>
int main() {
    printf("Hello World");
    return 0;
}
```

#include <stdio.h> provides standard input/output functions. **main()** is the starting point of a C program. **printf()** displays output, and **return 0** indicates successful completion.

3. Variables and Data Types

Common data types include **int** for integers, **float** for decimal values, **double** for higher-precision decimals, and **char** for a single character.

```
Example:
int age = 19;
float price = 99.50;
char grade = 'A';
```

4. Operators

Arithmetic operators: +, -, *, /, %

Relational operators: ==, !=, >, <, >=, <=

Logical operators: &&, ||, !

Assignment operators: =, +=, -=, *=, /=

5. Conditional Statements

The if, else if, and else statements are used to make decisions.

```
if (age >= 18) {
    printf("Adult");
} else {
    printf("Minor");
}
```

6. Loops

C provides for, while, and do-while loops for repeating statements.

```
for (int i = 1; i <= 5; i++) {  
    printf("%d\n", i);  
}
```

7. Functions

A function is a reusable block of code designed to perform a particular task. Functions help make programs easier to organize and maintain.

8. Arrays

An array stores multiple values of the same data type in contiguous memory.

```
int marks[5] = {80, 75, 91, 68, 88};
```

9. Pointers

A pointer is a variable that stores the memory address of another variable. The & operator obtains an address and * can be used to access the value at an address.

10. Important Exam Tips

- Understand the difference between = and ==.
- Remember that array indexing starts at 0.
- Use semicolons at the end of C statements.
- Practice if-else, loops, arrays, functions, and pointers.
- Compile and test small programs regularly.